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Model to Scale Rain Attenuation Time Series With Link Elevation Angle for LEO Satellite Based Systems

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Abstract A model to scale time series of rain attenuation along an Earth-to-satellite path with variable elevation angle is proposed. Starting from the time series at the lowest elevation the algorithm scales up each sample to the elevation of interest while maintaining the space and time correlation of the rain process and by correctly reproducing the local long-term rain attenuation statistics. Because of these peculiarities, the model represents a key element for the development of attenuation time series synthesizers for LEO satellite-based applications, where the time decorrelation of the rain event during the quick spacecraft passage over the ground station (roughly 10 min) is quite limited. Moreover, the comparison with the simple geometric scaling (i.e., the cosecant law) carried out among the attenuation Complementary Cumulative Distribution Functions shows a definite improvement in the prediction accuracy, even more so when the elevation angle falls below 20°.

Key Points:

- A model to scale rain attenuation time series with link elevation angle offering better accuracy over the customary cosecant law scaling
- The model generates a set of rain attenuation time series maintaining the time and space correlation as a function of link elevation
- The model's accuracy is evaluated in two sites with different climatic features

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1. Introduction

The last decades have been characterized by the explosion in the number of telecommunication (TLC) satellite systems in geosynchronous orbit (GSO) operating at frequencies higher than 10 GHz. Nowadays, the Ka band is largely used: by making available a larger bandwidth (about two times greater than at Ku band), it provides a much higher system capacity, as is required to cope with the increasing request of multimedia applications. To the same aim, the Q and V (and even the W) bands, are now under consideration for future applications (Fukuchi & Saito, 2007). At such high frequencies, extra path losses, mainly caused by the raindrop absorption and scattering mechanisms, increase steeply as the wavelength becomes closer to the rain drop size (Allnutt, 2011; Matricciani, 1996). The satellite design can no longer be approached by simply oversizing some system elements such as the antenna dimensions and the output power of the transmitters, but must be addressed with the use of one or more Fade Mitigation Techniques (FMTs), such as space or frequency diversity, on-board reconfigurable antenna and Adaptive Coding and Modulation.

For instance, new advanced TLC systems, such as the KA-SAT (Fenech et al., 2015), envisage the use of more than one gateway to efficiently cover a continental area. For the design of such systems, we need not only the long-term attenuation Complementary Cumulative Distribution Function (CCDF) at each ground station, $P(A)$, but also suitable information for dimensioning a large scale space diversity ground-based configuration in order to meet the availability and Quality of Service targets. While the first issue can be faced and solved with good accuracy using a $P(A)$ prediction method based on the knowledge of the local rain intensity CCDF (Capsoni et al., 2009; ITU-R P.618-13, 2017), as for the second issue, we must use a time series synthesizer for single site (Bertorelli et al., 2008; Lacoste et al., 2006) and/or multiple sites (Nebuloni et al., 2014).

A GSO satellite-based TLC system presents several advantages such as the permanent connection to the Earth station(s) and the possibility of providing global coverage with only three satellite units. Its design is carried out by following well-established procedures and by making use of prediction models with a solid background, which are continuously updated and improved (ITU-R P.618-13, 2017). Nevertheless, the GSO configuration also presents some drawbacks: it is not suitable for applications where, for instance, the length of the radio path to the satellite is challenging because of the limited output power of the ground (mobile) terminals (Al-Hraishawi et al., 2021) or in the special case of providing an efficient downlink for Earth observation data, EODDL (Arapoglou & Panagopoulos, 2011). Consequently, the use of non-GSO satellites is becoming more and more popular (Chen et al., 2021). For the design of non-GSO satellites we need, first of all, to predict long term rain attenuation CCDFs for all the sites of interest by taking in due account the peculiarity of the variable path elevation towards the satellite: this, in turn, translate in having reliable methods for scaling rain attenuation over the full range of elevations involved (Kourogiorgas et al., 2012; Pereira et al., 2014). In particular, when moving to

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elevation angles below roughly 20°, as it is the case for EODDL, the simple geometric scaling based on the variable path length in a homogeneous medium is no longer applicable (Tomaz & Capsoni, 2018).

This contribution presents a model, supported by some testing results, to generate a set of rain attenuation time series for any elevation above θ_{min}, whose ensemble preserves the real average time and space correlation of precipitation and mimic the long-term rain attenuation CCDFs. The model is intended as a first step towards the development of a complete time series generator specifically designed for LEO satellite links.

The paper is organized as follows: Section 2 briefly discusses and supports our choice of using the rain maps provided by the MultiEXCELL model as reference database for the model's development; Section 3 describes the procedure to obtain the elevation scaling factor from the rain attenuations CCDFs, $P(A,\theta)$, and derives its mathematical description at one geographical site; Section 4 shows how to extend the modeled scaling factor to the time series $A(\theta,t)$, which, all together, produce the $P(A,\theta)$; Section 5 presents the model results at two meteorologically different sites (in temperate and tropical regions, respectively) and the comparison with the simple geometric scaling. Finally, Section 6 draws some conclusions.

2. Reference Database

The propagation-oriented rain field model Multi EXponential CELL (MultiEXCELL) (Luini & Capsoni, 2011), whose reliability and worldwide applicability in different scenarios have been already extensively validated (Capsoni et al., 2010; Luini & Capsoni, 2010, 2013), was selected to provide us with the reference database. In fact, for modeling purposes, we were forced to use simulations rather than direct measurements because of the lack of available experiments where the attenuation values at different path elevations were jointly collected at the same site. The other option to use radar data is unfeasible in practice because it would imply obtaining from some external source a large database of raw radar data accurately calibrated for propagation purposes, and with the required spatial resolution, on which to apply a very cumbersome analysis aimed at deriving a statistically meaningful set of $A(\theta,t)$ time series.

First presented in Luini and Capsoni (2011), MultiEXCELL is a global rainfall model oriented to the analysis and prediction of the atmospheric radio propagation impairments. MultiEXCELL allows to generate a set of synthetic rain fields (an example is shown in Figure 1), whose ensemble preserves the local rainfall statistics (the only input to the model) and reproduces the correct rainfall spatial correlation.

The spatial distribution of rain rate R (mm/hr) in a single synthetic rain cell constituting the cells aggregate as shown in Figure 1, was first introduced by the EXponential CELL (EXCELL) model (Capsoni et al., 1987), for analytically reproducing the main properties of real rain cells. Specifically, the rain cell is modeled by the following equation:

$$R(\rho) = R_M e^{-\frac{\rho}{\rho_0}} \tag{1}$$

where ρ is the distance from the cell center, which is the place of the peak rain rate R_M, while ρ_0 is a parameter related to the rain cell dimension. EXCELL assumes that rain cells can be modeled worldwide an in Equation 1; what changes from site to site is the relative number of rain cells with different dimension and peak intensity: each cell, univocally defined by a (ρ_0,R_M) couple, is associated with a probability of occurrence $N(\rho_0,R_M)$, which in turn is calculated from the local rain rate CCDF, $P(R)$, using the following equation (Luini & Capsoni, 2011):

$$N(R_M,\rho_0) = -\frac{1}{\pi(\Psi_m - \Psi_M)\exp(2\sigma^2(R_M) + 2\mu(R_M))} \left.\frac{d^3P(R)}{d\ln(R)^3}\right|_{R=R_M} \cdot \frac{1}{\rho_0\sigma(R_M)\sqrt{2\pi}} \exp\left[-\left(\frac{\ln(\rho_0) - \mu(R_M)}{\sqrt{2}\sigma(R_M)}\right)^2\right] \tag{2}$$

where:

$$\Psi_m = \Psi\left[\frac{\mu(R_M) + 2\sigma^2(R_M) - \ln(a)}{\sqrt{2}\sigma(R_M)}\right] \text{ and}$$

$$\Psi_M = \Psi\left[\frac{\mu(R_M) + 2\sigma^2(R_M) - \ln(b)}{\sqrt{2}\sigma(R_M)}\right] \tag{3}$$

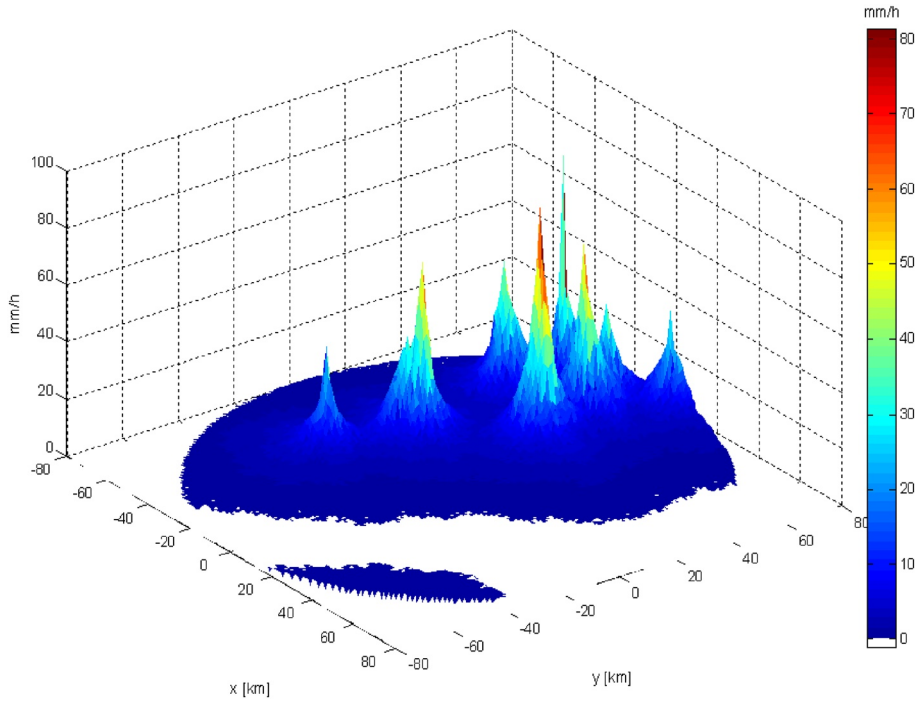


Figure 1. Example of the synthetic rain fields generated by the Multi EXponential CELL model.

$\Psi(x)$ is the error function:

$$\Psi(x) = \frac{2}{\sqrt{\pi}} \int_0^x \exp\left(-t^2\right) dt \tag{4}$$

whereas $\mu(R_M)$ and $\sigma(R_M)$ are given by:

$$\mu(R_M) = 7.86 R_M^{-0.84} - 6.03 R_M^{-17.27} + 0.001 R_M - 0.19 \tag{5}$$

$$\sigma(R_M) = 33.32 R_M^{-0.32} - 34.22 R_M^{-0.35} \tag{6}$$

MultiEXCELL synthetic rain fields, whose spatial resolution is 1×1 km^2 and whose dimension is 230×230 km^2, result from the arrangement of multiple synthetic exponential cells according to the natural rain cells' aggregative process observed in the real rain environment, and reproduce the statistical distribution of the rain coverage area (Luini & Capsoni, 2011). To this aim, the model includes a methodology that estimates, on a global basis, the distribution of the fractional area of a map affected by rain, starting from the European Centre for Medium-range Weather Forecast ERA-40 database (Uppala et al., 2005).

As a result, MultiEXCELL can be applied on a global basis and is a model suitable for diversified scenarios, from Earth-to-satellite communication systems with single (Capsoni et al., 2010) and multiple sites (Luini & Capsoni, 2013), to long terrestrial links (Luini & Capsoni, 2010), to the application considered here, where the knowledge of the large-scale spatial distribution of precipitation is of key importance: indeed, for low elevation angles, the portion of the link affected by rain is several kilometers long. Thanks to its mixed statistical-empirical nature, a key advantage of MultiEXCELL is that a relatively small set of synthetic rain fields (around 600 maps) is sufficient to reliably represent the local rainfall process and, thus, to allow the efficient simulation of the interaction between a telecommunication system and precipitation.

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Starting from the 600 synthetic rain maps generated for a given site, the attenuation affecting the Earth-space path is calculated numerically by using the standard Equation 7:

$$A_{\mathrm{dB}} = \int_0^{L_S} k[R(x,y)]^{\alpha} dL, \tag{7}$$

where: k and α are parameters dependent on frequency, polarization and elevation angle derived from Recommendation ITU-R P838-3 (ITU-R P.838-3, 2005); $R(x,y)$ is the rain rate in mm/hr drawn from the MultiEXCELL map and L_s is the path-length. The vertical profile of rain is assumed to be constant up to the rain height as suggested by Recommendation ITU-R P.839-4 (ITU-R P.839-4, 2013). Simulations were carried out from 5°, here assumed as the minimum operational elevation, to 90°, by applying the integration in Equation 7 at all the virtual ground stations set in each pixel of the rain maps (Tomaz et al., 2017): indeed, this approach allows increasing the number of samples and consequently achieving the required statistical stability of the results. Two meteorologically different sites were considered: Spino d'Adda, Italy (45.4°N, 9.5°E), in the temperate region, and Miami, USA (25.5°N, 80.3°W), in the tropical region. Simulations at Spino d'Adda were carried out at 39.6 GHz with circular polarization because the local $P(A)$ measured during the ITALSAT experiment was used to validate the procedure (Polonio & Riva, 1998). For Miami instead, we considered the lower 27 GHz frequency, first because it is in the band assigned to LEO applications and also because Miami has marked convective characteristics, and, as such, the attenuation values are extremely high.

An indirect proof that MultiEXCELL allows reproducing realistic rain maps, reflecting the local meteorology, is confirmed by the accurate results presented in Luini and Capsoni (2013), where the model was used to predict the gain in attenuation for site diversity systems. Tests were conducted against all the measurements included in the database of two-site diversity experiments maintained by the ITU-R Study Group 3 (DBSG3), which includes data collected in several countries characterized by different climates, from tropical to temperate (latitude ranging from 27.6°N for Tampa, FL, USA, to 55.7°N for Albertslund, Denmark).

3. Derivation of the Scaling Factor

The high variability of the precipitation process is a very complex feature to model because at a given time t_0 the attenuation values at different elevations, $A(t_0,\theta)$, depend not only on the rainy path-length along the link (not related deterministically to the path elevation) but also on the actual rain profile (variable both in time and in space). This necessarily implies that the use of a model for $A(t,\theta)$ with a sound physical basis (such as MultiEXCELL) is not achievable but a statistical approach is the only possible solution. As shown in Equation 8, we have described the complex variability of $A(t,\theta)$ as the superposition of a long term (average) component and a variable one (Δ), respectively associated with the spatial variability of rain intensity along the path and to the variable length and rain profile of the radio path.

$$A(t,\theta) = < A(t,\theta) > + \Delta(t,\theta) \tag{8}$$

Evidence of the above behavior was observed in the attenuation variation with θ provided by MultiEXCELL simulations.

The present contribution will address only the average component that is of direct interest for the statistical description of the behavior of a LEO link. As for the variable component (Δ), a very preliminary analysis showed that its spread is not negligible and can reach, within a rain event, deviations as large as ±40% in dB, with few cases exceeding this value. The variable component contribution to the overall database is zero, but it not negligible on an event basis. Some events are overestimated by the long-term value, while some others are underestimated. The "true" pattern of $A(t,\theta)$ during the satellite passage over an Earth station is of interest in specific applications as, for instance, when testing the efficiency of FMTs.

The basic idea underlying the model's development of the average component $<A(t,\theta)>$ is that the long-term rain attenuation CCDF, $P(A,\theta)$, is the result of the sum of all the attenuation events occurring in multiple years and, as a consequence, there is room to consider that the scaling factor of $P(A,\theta)$ could be used for the time series average components as well. This speculation has proven to be correct and is detailed in the following of this contribution.

Several $P(A,\theta)$ were derived from the database as per Section 2, where $P_0(\theta)$ represents the probability of attenuation greater than zero for each elevation angle (see Figure 2). As expected, $P_0(\theta)$ decreases as the elevation

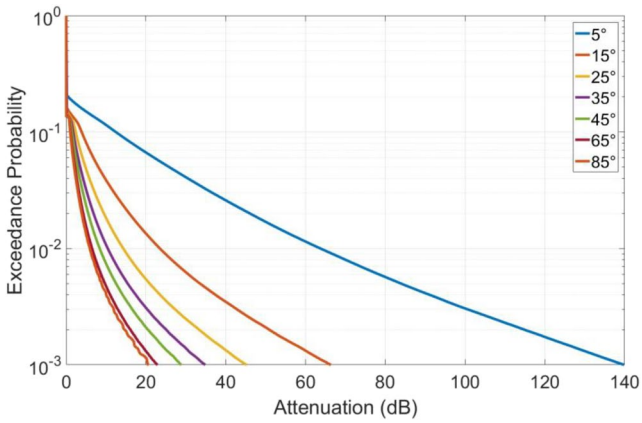


Figure 2. Sample of the database of rain attenuation Complementary Cumulative Distribution Functions obtained through the procedure described in Section 2. Spino d'Adda, 39.6 GHz circular polarization.

angle increases because it tightly depends on the length of the rainy path (a longer length means a higher probability of incercepting rain as clearly shown in Figure 3). It is also worth pointing out that there is a huge increase in the attenuation, for any exceedance probability, for elevation angles lower than 20°: indeed, this is a sign that the radio path is affected by more than one rain cell.

In order to avoid introducing any dependence of the model's parameters on the probability of rain occurrence that, in general, differs from site to site, we introduced a normalization so as to obtain $P_0^*(5°) = 1$ (i.e., $P_0^*(\theta) = P_0(\theta)/P_0(5°)$). The results are shown in Figure 3. The asterisk superscript indicates the quantities after normalization. In addition to the parameters of the normalized CCDF $P^*(A\theta)$, the scaling factor depends on the rain height h_r, that is, the other parameter influencing the rainy path length.

The analysis of simulations relative to the Spino d'Adda site (Italy) has shown that $P_0^*(\theta)$ can be modeled as in Equation 9, with θ varying from 6° to 90°. The a, b, and c coefficients explicitly depend on h_r (km) as indicated in Equation 10. All the other values $A_{\mathrm{dB}}(\theta,P^*)$ are related to one other through the scaling factor $\phi(\theta,P^*)$ reported in Equation 11. Please note that, throughout this contribution, the ^ sign is used to define the estimator of the quantity indicted beneath the sign itself.

$$\hat{P}_0^*(\theta) = a[\sin(5°)\mathrm{cosec}(\theta)]^b + c \tag{9}$$

$$\begin{cases} a = 0.0714h_r + 0.2159 \\ b = 0.01455h_r + 0.6499 \\ c = -0.0711h_r + 0.7686 \end{cases} \tag{10}$$

$$\phi(\theta, P^*) = \frac{A_{\mathrm{dB}}(\theta, P^*)}{A_{dB}(5°, P^*)\sin(5°)\mathrm{cosec}(\theta)} \tag{11}$$

In Equation 11, P^* is the normalized probability level of A_{dB}. Note that $\phi(\theta,P^*)$ acts de facto as a correction factor for the geometric scaling.

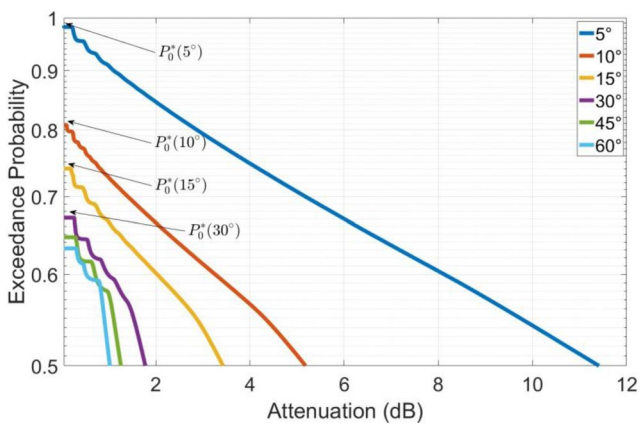


Figure 3. Complementary Cumulative Distribution Functions normalized to the 5° elevation, zoom, 39.6 GHz circular polarization.

Figure 4 shows the plot of the scaling factor ϕ for some sample elevations as a function of P^*. For an appropriate modeling of the curves, it is evident the need to subdivide the probability scale in intervals and use different analytical functions for each one of them. Specifically, to model $\phi(\theta, P^*)$ as $\hat{\phi}(\theta, P^*)$, we considered two main intervals, the one on the right and the one on the left of the vertical dotted line in Figure 4, respectively. For:

$0.1 \le P^* < P_0^*(\theta)$

We used the second-order polynomial function in Equation 12, but as the probability $P_0^*(\theta)$ changes with elevation in a complex way, as it can be easily appreciated from the right side of Figure 4, we further subdivided the interval in two parts, in order to obtain the best fitting accuracy over the entire probability range. For $P^* \ge 0.5$, the parameters α_1, α_2, α_3, in Equation 12 vary linearly with θ (see Equation 13), while for $P^* < 0.5$, their dependence is exponential (see Equation 14).

$$\hat{\phi}(\theta, P^*) = \alpha_1(P^*)^2 + \alpha_2 P^* + \alpha_3 \tag{12}$$

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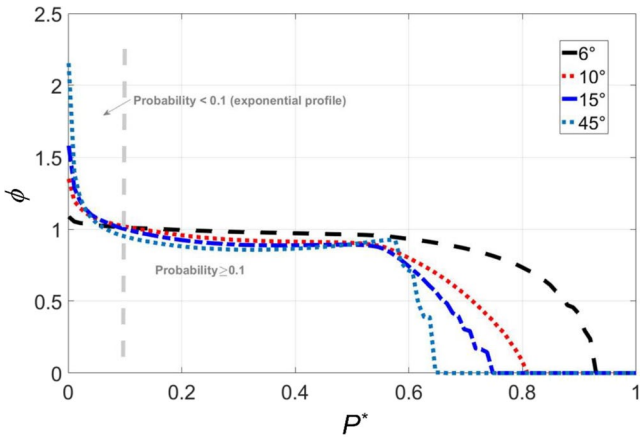


Figure 4. Scaling factor ϕ as a function of the normalized exceedance probability P^*.

Accordingly, the parameters α_1, α_2, and α_3 are:

$$\begin{cases} \alpha_1 = -1.361\theta + 5.797 \\ \alpha_2 = 1.483\theta - 7.394 \quad \text{for} \quad P^* \geq 0.5 \\ \alpha_3 = -0.0711 h_r + 0.7686 \end{cases} \tag{13}$$

$$\begin{cases} \alpha_1 = a_1 e^{b_1\theta} + c_1 e^{d_1\theta} \\ \alpha_2 = a_2 e^{b_2\theta} + c_2 e^{d_2\theta} \quad \text{for} \quad P^* < 0.5 \\ \alpha_3 = a_3 e^{b_3\theta} + c_3 e^{d_3\theta} \end{cases} \tag{14}$$

As for Equation 14, the coefficients a_j, b_j, c_j, d_j, for $j = 1, 2, 3$, are listed in Table 1.

$P^* < 0.1$

In this case, the scaling factor $\hat{\phi}(\theta, P^*)$ follows an exponential behavior for any elevation as indicated in Equation 15, where the parameters β_1 and β_2 are given in Equations 16 and 17, respectively.

$$\hat{\phi}(\theta, P^*) = \beta_1 e^{\beta_2 \log(P^*)} \tag{15}$$

$$\beta_1 = 0.5428 e^{-0.08142\theta} + 0.6473 e^{-0.00108\theta} \tag{16}$$

$$\beta_2 = -0.3817 e^{0.002311\theta} + 0.5507 e^{-0.07894\theta} \tag{17}$$

Note that the largest difference among the curves depicted in Figure 4 is the higher range of probability where the curves behavior is highly influenced by their $P_0^*(\theta)$ value (see Figure 3).

4. Time Series Scaling

As mentioned in the previous section, the scaling factor $\hat{\phi}(\theta, P^*)$ can also be used to scale each sample of the rain attenuation time series. In this regard, the best procedure we devised is to generate the latter by starting from the lowest elevation θ_{min} and moving up with θ (upscaling). Indeed, although the choice of a generic elevation θ_a as a starting angle would be attractive if direct rain attenuation measurements at θ_a from a GSO satellite are available, it requires both up and down scaling to cover the full θ range: in practice, this procedure has shown to yield less accurate predictions. The ensemble $A(\theta_{min},t)$ to be used as input to the model for a given site can be obtained:

- from experiments (although direct measurements at very low θ are collected only by Earth-stations located at high latitudes);
- from models, that is, from the class of rain attenuation time series synthesizers able to provide a reliable low elevation data set.

In this regard, we have selected the time series generator proposed in (Bertorelli et al., 2008) because of its physical sound basis and the proven very good performance. According to Bertorelli et al. (2008), any $P(A)$ can be reconstructed with high accuracy by summing up a weighted set of attenuation time series sorted from a large database of measured ones (ITASAT rain attenuation measurements collected over many years), according to their peak value. The original version was modified so as to extend its applicability down to very low θ. Specifically, the modification consists in increasing the original database with a set of synthetic events resulting from the summation of two out of those that are part of the original database. This procedure is intended to reproduce the physical situation occurring at low elevations

Table 1
Coefficients in Equation 14

	a_j	b_j	c_j	d_j
$j = 1$	1.656	0.0002381	−6.32	−0.235
$j = 2$	−1.156	−0.0007235	6.82	−0.3262
$j = 3$ for $\theta \leq 12°$	1.259	−0.008586	−0.6954	−0.2412
$j = 3$ for $\theta > 12°$	0.1539	−0.05058	1.018	5.767×10^{-5}

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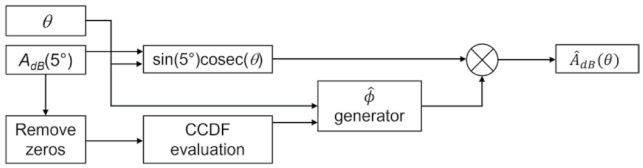


Figure 5. Model diagram.

where the link crosses more than one rain structure. The resulting database is then enlarged by 20 times through the summation of two episodes (rain events), that are randomly shifted one with respect to the other. More details on the procedure are given in Tomaz et al. (2018).

Figure 5 shows a schematic block diagram illustrating the elevation scaling procedure for the attenuation time series. The block "$\hat{\phi}$ generator" produces a value $\hat{\phi}$ for each sample. This value comes from the implementation of Equations 12 and 15 according to the probability value P^* of the sample provided by the block "CCDF evaluation." If the probability of the sample is greater than $\hat{P}_0^*(\theta)$, then the value of $\hat{\phi}$ is zero. The following step consists in multiplying each input sample previously scaled with the cosecant law, as in Equation 18, by the factor $\hat{\phi}(\theta, P^*)$. Indeed, $\hat{\phi}(\theta, P^*)$ acts as a correction factor of the simple geometric scaling.

$$\hat{A}_{\mathrm{dB}}(\theta, n) = A_{dB}(5°, n)[\sin(5°)\mathrm{cosec}(\theta)]\hat{\phi}(\theta, P^*) \tag{18}$$

In Equation 18, n is the sequential sample number in the time series.

5. Results

The evaluation of the scaling accuracy is performed on the basis of the figure of merit ε proposed in Recommendation ITU-R P.311-15 (ITU-R P.311-15, 2015)—see Equation 19, and specifically on its root mean square (RMS) value reported in Equation 20 for convenience. The ε value is calculated for the following standard ITU-R probability levels: (0.5 0.4 0.3 0.2 0.1 0.05 0.03 0.02 0.01 0.005 0.003 0.002 0.001).

$$\begin{cases} \varepsilon = 100\left[\dfrac{A_m(P)}{10}\right]^{0.2}\log\left[\dfrac{A_{\mathrm{sc}}(P)}{A_m(P)}\right] & \text{for} \quad A_m < 10\,\mathrm{dB} \\ \varepsilon = 100\log\left[\dfrac{A_{\mathrm{sc}}(P)}{A_m(P)}\right] & \text{for} \quad A_m \geq 10\,\mathrm{dB} \end{cases} \tag{19}$$

$$\mathrm{RMS}(\varepsilon) = \sqrt{\mu_\varepsilon^2 + \sigma_\varepsilon^2} \tag{20}$$

In Equation 19, A_{sc} is the scaled attenuation (using the model or the simple cosecant scaling) and A_m is the reference attenuation obtained from simulations on the MultiEXCELL maps; μ_ε and σ_ε are, espectively, the mean and the standard deviation of ε.

Figure 6 shows the CCDFs of the normalized synthetic time series for Spino d'Adda at 5° and the one scaled to 90° (for reference, the $P = 1$ level in Figure 6 corresponds to about 5×10^{-2} of the absolute time, i.e., the mean yearly probability to have rain attenuation in Spino D'Adda). Note how well the curve scaled with the proposed

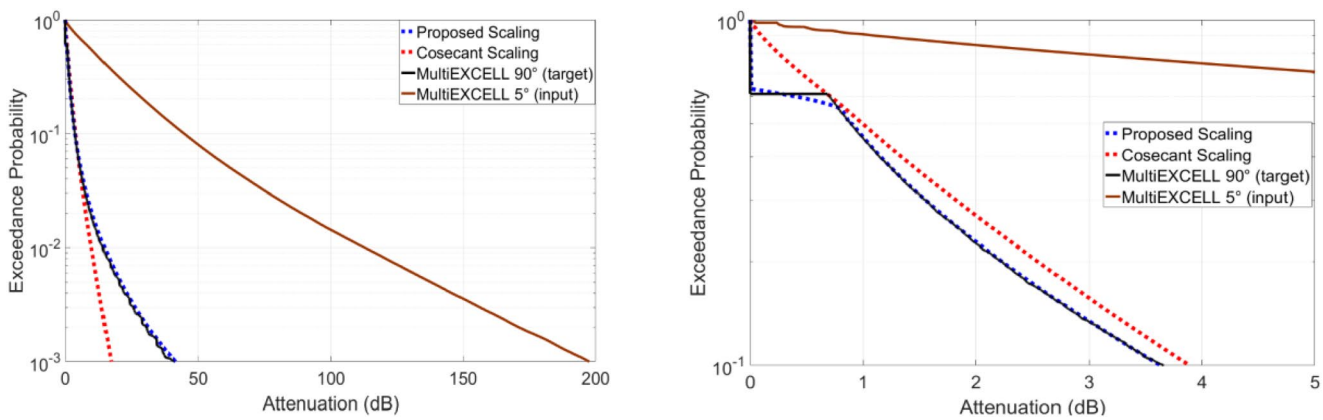


Figure 6. Complementary Cumulative Distribution Function of the time series scaled from 5 to 90 for Spino d'Adda, 39.6 GHz, circular polarization: full picture on the left side; high probability detail on the right side.

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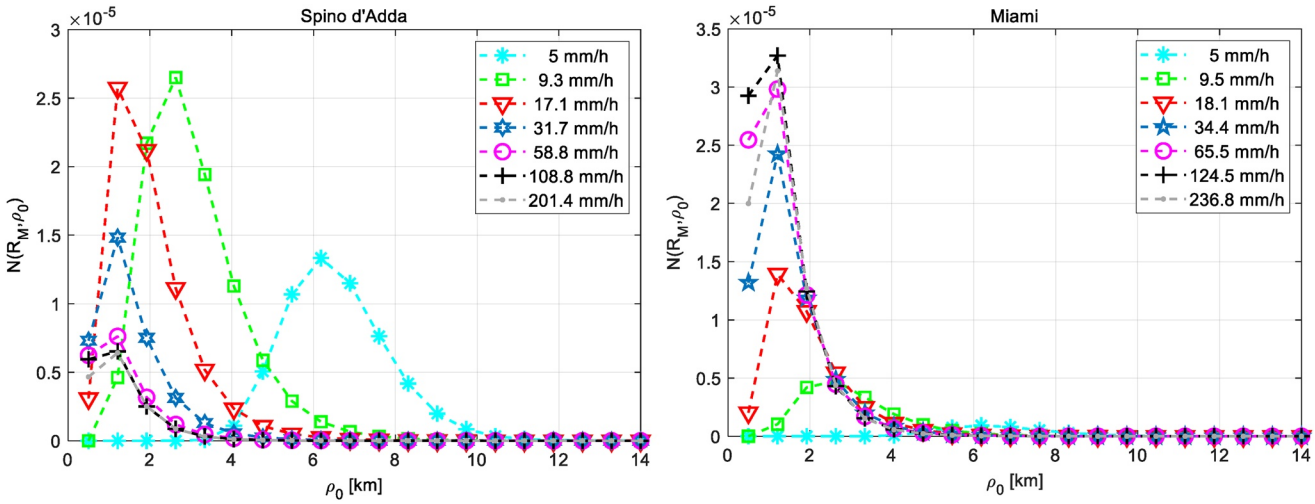


Figure 7. Rain cells' probability of occurrence calculated using Equation 2 for Spino d'Adda (left side) and Miami (right side).

model matches the target one even when considering the largest elevation difference (RMS(ε) = 3.5 for the model, and RMS(ε) = 41.1 for the cosecant scaling). Specifically, the 5° elevation curve (solid red line) was derived from MultiEXCELL maps while the 90° one (blu dotted line) was obtained from the time series scaling model here in this work. Finally, the reference 90° curve (solid black line) was simply obtained by converting each rain rate value of the local $P(R)$, input to MultiEXCELL, to the correspondent specific attenuation and then by multiplying it by the local rain height h_r (no change of the rain rate along the zenithal direction). In mathematical terms:

$$A_{\mathrm{dB}}(90°, P^*) = kR(P^*)^{\alpha} h_r \tag{21}$$

In order to verify to what extent the scaling technique is applicable to different sites, using the same coefficients proposed in Section 3, we considered Miami USA (tropical climate), where the number of convective events is definitely larger than in Spino d'Adda (temperate climate): this is clearly visible by comparing the rain cells' probability of occurrence N in Figure 7 calculated using Equation 2 for the two sites. The two figures clearly evidence the huge difference between the two sites: in Miami the occurrence probability for cells with low peak rain is almost negligible, while the ones with more intense peaks (even exceeding 120 mm/hr) and very limited in spatial extent are associated to higher occurrence. Moreover, the statistical distribution of the rain coverage area is totally different. Finally, the mean yearly rain height is significantly higher: 3.34 and 4.59 km for Spino d'Adda and Miami, respectively, as per Recommendation ITU-R P.839-4 (ITU-R P.839-4, 2013).

The results are shown in Figure 8 where the origin of the curves is the same of the curves in Figure 6. Again, the agreement is indeed very good despite the coefficients of the model are the ones derived for Spino d'Adda. For comparison, Figure 9 shows the RMS and mean values of the figure of merit obtained for Miami from the model and the cosecant scaling.

As a last result, Figure 10 shows an example of the rain attenuation event generated with the procedure presented in (Tomaz & Capsoni, 2018), (Spino d'Adda, 39.6 GHz, at 5°) and scaled with the model proposed in this contribution to 7°, 15°, and 90°. Note that the scaled event is not only very reduced in terms of amplitude but is also shorter in terms of duration (reference: the zero dB level), as it should be. Moreover, the peaks of the event are better preserved by this model (continuous lines) rather than by the customary cosecant scaling (dotted lines).

6. Conclusions

A model to generate a statistically meaningful set of rain attenuation time series as a function of link elevation, able to maintain the real average time and space correlation of precipitation is here presented and discussed. The complex variability of $A(t,\theta)$ has been described by the superposition of a long term (average) component and a variable one, respectively associated with the spatial variability of the rain intensity along the path and to the variable length and rain profile of the radio path. Only the average component of the process is addressed in the

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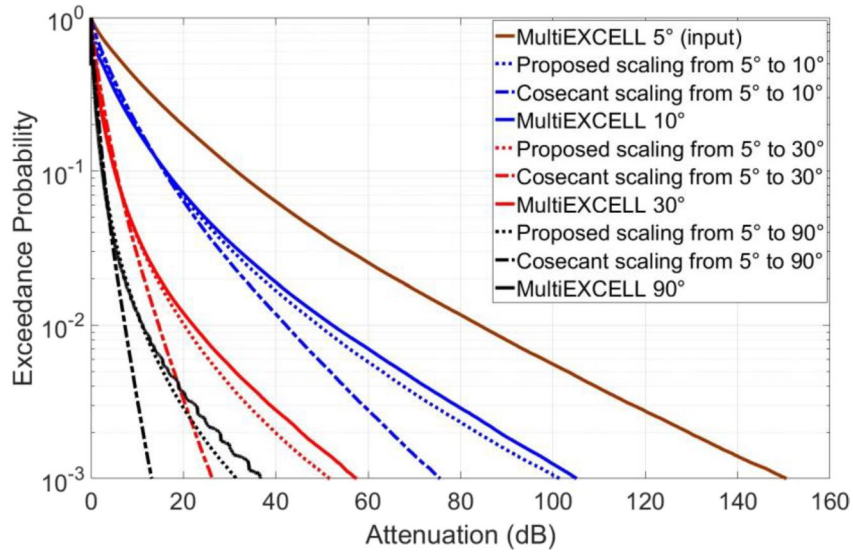


Figure 8. Scaling procedure applied to the data obtained for Miami 27 GHz, circular polarization, for some elevation angles.

paper. This model is intended as a first step toward the development of a complete time series generator specific for LEO satellite link, where the time decorrelation of the rain event during the quick spacecraft passage (roughly 10 min) is quite limited. Results show how the proposed model performs very well when compared in terms of resulting attenuation CCDFs with the reference (MultiEXCELL simulations). First tests give also confidence on its applicability to sites with very different meteorological characteristics. The worldwide application of the model will require the evaluation of $\hat{\phi}(\theta, P^*)$ in some selected sites with different meteorological characteristics and the computation of a "site-average" set of parameters α_i and β_i. Further analysis should be done in order to verify the ability of the model to adequately reproduce rain fade dynamics parameters such as fade slope and fade duration. This requirement must be met to ensure that the second order statistics of the rain process are reproduced as well. In particular this test could be limited to the lower elevations, because at the higher ones, it is already "verified" due to the peculiarity of the original time series generator (Bertorelli et al., 2008) based on the measured rain attenuation time series (ITALSAT).

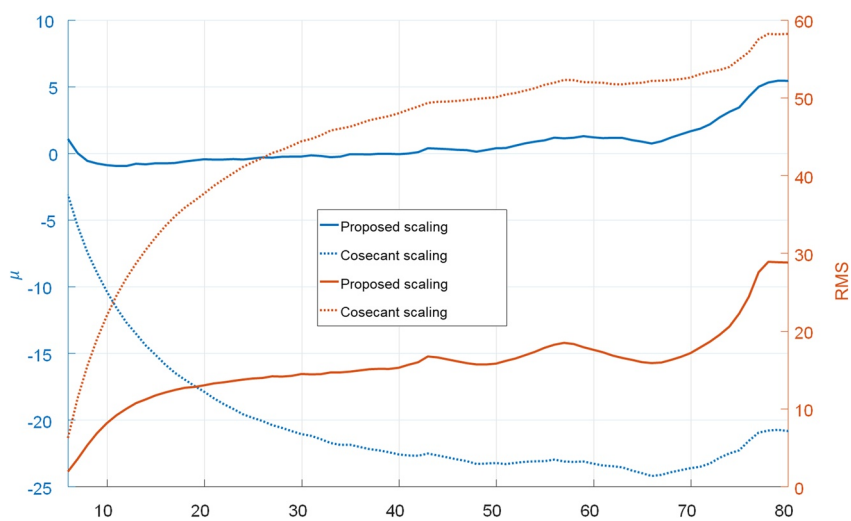


Figure 9. Mean value and root mean square value of the figure of merit evaluated from 6° to 80° elevation from Miami, 27 GHz, circular polarization.

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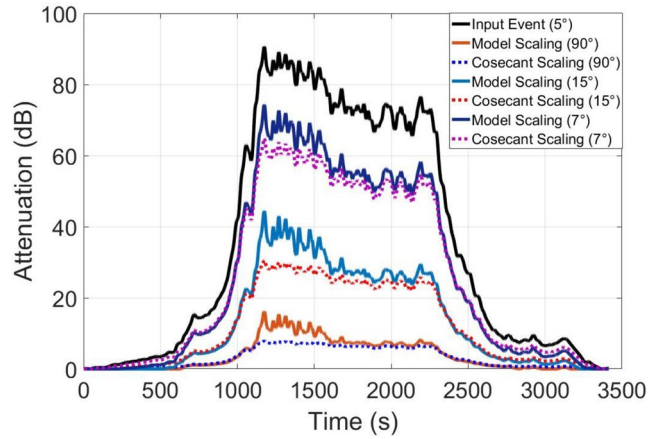


Figure 10. Example of rain event generated by the method proposed in Tomaz and Capsoni (2018) and scaled from 5° to 7°, 15°, and 90° for Spino d'Adda, 39.6 GHz.

Data Availability Statement

The MultiEXCELL synthetic maps used for the model development are available in the following public repository: https://zenodo.org/record/7504219/files/MultiEXCELL.rar?download=1. The ITALSAT data used to derive the time series database are not accessible to the public or research community as they were collected in the framework of the ITALSAT experiment funded by the Italian Space Agengy. For information on the chance to access the data, user can contact the following address: carlo.riva@polimi.it.

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